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EMPLOYMENT

2018-	University of Virginia , Associate Professor <i>School of Data Science</i> <i>Dept. of Systems and Information Engineering</i>
2022-2024	Amazon , Amazon Scholar
2018-2019	University of Virginia , Associate Professor <i>Darden School of Business</i>
2013-2018	University of Alabama , Assistant/Associate Professor <i>Information Systems, Statistics, Management Science</i>
2008-2013	Spadac/GeoEye/DigitalGlobe , Principal Research Scientist
2007-2008	Statistical and Applied Mathematical Sciences Institute (SAMSI) , Post-doc
2006-2008	North Carolina State University , VIGRE Postdoc Fellow <i>Department of Statistics</i>
1998-2001	Sanford/Newell-Rubbermaid , Project Engineer

EDUCATION

2006	Ph.D.	University of Virginia	Systems and Information Engineering
2003	M.S.	Vanderbilt University	Systems Engineering/Mgmt of Technology
1998	B.S.	Purdue University	Industrial Engineering

PUBLICATIONS

1. Rostami-Tabar, B., Pinson, P., and Porter, M. D. “Guest editorial: Forecasting for social good”. In: *International Journal of Forecasting* 41.1 (2025), pp. 1–2
2. Tian, J. and Porter, M. D. “Deep-Learning Based Probabilistic Forecasting Framework for Censored Data”. In: *23rd IEEE International Conference on Machine Learning and Applications (ICMLA)*. 2024
3. Pennetti, C. A. and Porter, M. D. “AI in Motion”. In: *Civil Engineering Magazine* 94.6 (2024), pp. 60–69
4. Liu, H., Tang, X., Chen, T., Liu, J., Indu, I., Zou, H. P., Dai, P., Galan, R. F., Porter, M. D., Jia, D., Zhang, N., and Xiong, L. “Sequential LLM Framework for Fashion Recommendation”. In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Miami, Florida, US: Association for Computational Linguistics, Nov. 2024, pp. 1276–1285
5. Tian, J. and Porter, M. D. “Time of week intensity estimation from partly interval censored data with applications to police patrol planning”. In: *Journal of Applied Statistics* (2024), pp. 1–19
6. Mostafavi, M., Porter, M. D., and Robinson, D. T. “Contextual Embeddings in Sociological Research: Expanding the Analysis of Sentiment and Social Dynamics”. In: *Sociological Methodology* (2024), p. 00811750241260729

7. McCulloch, M., Alonzi, L., White, S., Haregu, F., and Porter, M. "Pediatric donor heart acceptance practices in the United States: What is really being considered?" In: *Pediatric Transplantation* 28.1 (2024), e14649
8. Ruggeri, F., Porter, M. D., and White, G. "Endogenous and exogenous effects in self-exciting process models of terrorist activity". In: *Statistica Neerlandica* 79.1 (2025), e12347
9. Deverill, H., Scherer, W., Porter, M., and Stam, A. "The Utility of Machine Learning Applied to Military Assessment and Selection". In: *Military Operations Research* 29.2 (2024), pp. 53–94
10. Haregu, F., Dixon, R. J., Porter, M., and McCulloch, M. "Pediatric donor heart utilization variability among organ procurement organizations". In: *Pediatric Transplantation* 28.3 (2024), e14747
11. Haregu, F., Dixon, R., Porter, M., Scharff, J., and McCulloch, M. "Machine Learning for Predicting Waitlist Survival in Pediatric Patients Awaiting Heart Transplantation". In: *The Journal of Heart and Lung Transplantation* 43.4 (2024), S621–S622
12. Porter, M., Sharff, J., Dixon, R., Haregu, F., and McCulloch, M. "Using Machine Learning to Assess the Predictive Power of Donor Characteristics in Pediatric Heart Transplant Outcomes". In: *The Journal of Heart and Lung Transplantation* 43.4 (2024), S622
13. Rogers, A., Dibsie, C., Kuzneski, E., Underwood, D., Brozey, R., Sullivan, L., Robinson, D., and Porter, M. D. "Forecasting Breakthroughs: Identifying Future Leaders in the Semiconductor Industry". In: *2024 Systems and Information Engineering Design Symposium (SIEDS)*. IEEE. 2024, pp. 425–430
14. Jiang, Y. and Porter, M. D. "Quantifying the Influence of User Behaviors on the Dissemination of Fake News on Twitter with Multivariate Hawkes Processes". In: *arXiv preprint arXiv:2308.13927* (2023)
15. Haregu, F., Porter, M., Dixon, J., and McCulloch, M. "Effect of Practice Variation Amongst Organ Procurement Organizations on Pediatric Donor Heart Utilization". In: *The Journal of Heart and Lung Transplantation* 42.4 (2023), S446–S447
16. Mostafavi, M. and Porter, M. D. "Reevaluating Data Partitioning for Emotion Detection in EmoWOZ". in: *arXiv preprint arXiv:2303.13364* (2023)
17. Guleria, S., Schwartz, B., Sharma, Y., Fernandes, P., Jablonski, J., Adewole, S., Srivastava, S., Rhoads, F., Porter, M., Yeghyayan, M., et al. "The intersection of video capsule endoscopy and artificial intelligence: addressing unique challenges using machine learning". In: *arXiv preprint arXiv:2308.13035* (2023)
18. Moghadasi, N., Piran, M., Baek, S., Valdez, R. S., Porter, M. D., Johnson, D., and Lambert, J. H. "Systems Analysis of Bias and Risk in AI-Enabled Medical Diagnosis". In: *2023 IEEE Symposium Series on Computational Intelligence (SSCI)*. IEEE. 2023, pp. 1800–1807
19. Tian, J. and Porter, M. D. "Changing presidential approval: Detecting and understanding change points in interval censored polling data". In: *Stat* 11.1 (2022), e463
20. Jiang, Y. and Porter, M. D. "Simulating Fake News Dissemination on Twitter with Multivariate Hawkes Processes". In: *2022 IEEE International Conference on Big Data (Big Data)*. IEEE. 2022, pp. 3597–3606
21. Kotay, S. M., Tanabe, K. O., Colosi, L. M., Poulter, M. D., Barry, K. E., Holstege, C. P., Mathers, A. J., and Porter, M. D. "Building-Level Wastewater Surveillance for SARS-CoV-2 in Occupied University Dormitories as an Outbreak Forecasting Tool One Year Case Study". In: *ACS ES&T Water* 2.11 (2022), pp. 2094–2104
22. Rostami-Tabar, B., Hong, T., and Porter, M. D. "Guest Editorial: Forecasting for social good". In: *International Journal of Forecasting* 38.3 (2022), pp. 1173–1174

23. McCulloch, M., Liu, I., Alonzi, L., White, S., and Porter, M. "Assessing the Relationship Between Pediatric Donors Terminal Hospitalizations and Heart Acceptance Practices". In: *The Journal of Heart and Lung Transplantation* 41.4 (2022), S508
24. Mostafavi, M., Porter, M. D., and Robinson, D. T. "Learning affective meanings that derives the social behavior using Bidirectional Encoder Representations from Transformers". In: *arXiv preprint arXiv:2202.00065* (2022)
25. Rostami-Tabar, B., Ali, M. M., Hong, T., Hyndman, R. J., Porter, M. D., and Syntetos, A. "Forecasting for social good". In: *International Journal of Forecasting* 38.3 (2022), pp. 1245–1257. eprint: <https://arxiv.org/abs/2009.11669>
26. Jablonski, J., Fernandes, P., Adewole, S., Syed, S., Brown, D., and Porter, M. "Multi-frame Abnormality Detection in Video Capsule Endoscopy". In: *Proceedings of the Future Technologies Conference*. Vol. 2. Springer. 2022, pp. 177–186
27. Mohler, G. and Porter, M. D. "A note on the multiplicative fairness score in the NIJ recidivism forecasting challenge". In: *Crime Science* 10.17 (2021), pp. 1–5
28. Colosi, L. M., Barry, K. E., Kotay, S. M., Porter, M. D., Poulter, M. D., Ratliff, C., Simmons, W., Steinberg, L. I., Wilson, D. D., Morse, R., Zmick, P., and Mathers, A. J. "Development of Wastewater Pooled Surveillance of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) from Congregate Living Settings". In: *Applied and Environmental Microbiology* 87.13 (2021), e00433–21
29. Adewole, S., Fernandes, P., Jablonski, J., Copland, A., Porter, M., Syed, S., and Brown, D. "Graph Convolutional Neural Network For Weakly Supervised Abnormality Localization In Long Capsule Endoscopy Videos". In: *2021 IEEE International Conference on Big Data (Big Data)*. IEEE. 2021, pp. 388–399
30. Fazli, M., Sklar, S., Porter, M. D., French, B. A., and Shakeri, H. "Wastewater-Based Epidemiological Modeling for Continuous Surveillance of COVID-19 Outbreak". In: *2021 IEEE International Conference on Big Data (Big Data)*. IEEE. 2021, pp. 4342–4349
31. Mostafavi, M. and Porter, M. D. "How emoji and word embedding helps to unveil emotional transitions from social media interactions". In: *IEEE International Systems Conference (SysCon)*. 2021
32. Mostafavi, M., Porter, M. D., Jiang, Y., Phillips, M., and Freedman, P. "A Tale of Two Metrics: Polling and Financial Contributions as a Measure of Performance". In: *IEEE International Systems Conference (SysCon)*. 2021
33. Jiang, Y. and Porter, M. D. "Discovering Influence of Yelp Reviews Using Hawkes Point Processes". In: *Intelligent Systems Conference (IntelliSys)*. 2021
34. Adewole, S., Fernandes, P., Jablonski, J., Copland, A., Porter, M., Syed, S., and Brown, D. "Lesion2Vec: Deep Meta Learning for Few-Shot Lesion Recognition in Capsule Endoscopy Video". In: *Proceedings of the Future Technologies Conference*. Springer. 2021, pp. 762–774
35. Bullock, J., Grieco, M., Liu, Y., Pedersen, I., Roberson, W., Wright, G., Alonzi, P., McCulloch, M. A., and Porter, M. D. "Determining Factors of Heart Quality and Donor Acceptance in Pediatric Heart Transplants". In: *Systems and Information Engineering Design Symposium (SIEDS)*. 2021
36. Gresham, T. R., Kim, J., McDonald, J., Scoggins, N., Mostafavi, M., Park, B. B., Porter, M. D., Duffy, M. E., and Smith, S. A. "Safe and Sustainable Fleet Management with Data Analytics and Training". In: *Systems and Information Engineering Design Symposium (SIEDS)*. IEEE. 2021, pp. 1–3
37. Jiang, S., Maggard, K., Shakeri, H., and Porter, M. D. "An Application of the Partially Observed Markov Process in the Analysis of Transmission Dynamics of COVID-19 via Wastewater". In: *Systems and Information Engineering Design Symposium (SIEDS)*. 2021

38. Mohler, G., Porter, M., Carter, J., and LaFree, G. "Learning to rank spatio-temporal event hotspots". In: *Crime Science* 9.3 (2020), pp. 1–12
39. Campbell, E., Chamberlayne, E., Gawrylowicz, J., Hood, C., Hudak, A., Orlowsky, M., Rivero, E., and Porter, M. "Optimization of VDOT Safety Service Patrols to Improve VDOT Response to Incidents". In: *Systems and Information Engineering Design Symposium (SIEDS)*. IEEE. 2020, pp. 1–6
40. Koch, A., Tian, J., and Porter, M. D. "Criminal Consistency and Distinctiveness". In: *Systems and Information Engineering Design Symposium (SIEDS)*. IEEE. 2020, pp. 1–3
41. Pennetti, C. A., Andrews, D., Porter, M. D., and Lambert, J. H. "Traveler Perception of Transportation System Performance Using Kernel Density Estimation to Prioritize Infrastructure Investments". In: *International Conference on Transportation and Development*. 2020, pp. 48–61. eprint: <https://ascelibrary.org/doi/pdf/10.1061/9780784483169.005>
42. Cheng, F., Porter, M. D., and Colosi, L. M. "Is hydrothermal treatment coupled with carbon capture and storage an energy-producing negative emissions technology?" In: *Energy Conversion and Management* 203.112252 (2020)
43. Porter, M. D. and Akakpo, A. "Detecting, identifying, and localizing radiological material in urban environments using scan statistics". In: *IEEE International Symposium on Technologies for Homeland Security (HST)*. IEEE. 2019, pp. 1–6. eprint: <https://arxiv.org/abs/2002.03224>
44. Abrisqueta, A., Bishop, C. A., Perryman, S. P., Shoebbotham, L. M., Wang, J., and Porter, M. "Evaluation of VDOT's Safety Service Patrols to Improve Response to Incidents". In: *Systems and Information Engineering Design Symposium (SIEDS)*. IEEE. 2019, pp. 1–5
45. Mohler, G. and Porter, M. "Rotational grid, PAI-maximizing crime forecasts". In: *Statistical Analysis and Data Mining* 11.5 (2018), pp. 227–236
46. Wang, K. and Porter, M. "Optimal Bayesian Clustering using Non-negative Matrix Factorization". In: *Computational Statistics and Data Analysis* 128 (2018), pp. 395–411
47. Wang, K., Simandl, J., Porter, M., Graettinger, A., and Smith, R. "How the Choice of Safety Performance Function Affects the Identification of Important Crash Prediction Variables". In: *Accident Analysis and Prevention* 88.1 (2016), pp. 1–8
48. Porter, M. "A Statistical Approach to Crime Linkage". In: *The American Statistician* 70.2 (2016), pp. 152–165
49. Bouhana, N., Johnson, S. D., and Porter, M. D. "Consistency and specificity in burglars who commit prolific residential burglary: Testing the core assumptions underpinning behavioural crime linkage". In: *Legal and Criminological Psychology* 21.1 (2016), pp. 77–94
50. Reich, B. and Porter, M. "Partially-supervised spatiotemporal clustering for burglary crime series identification". In: *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 178.2 (2015), pp. 465–780
51. White, G. and Porter, M. "GPU accelerated MCMC for modeling terrorist activity". In: *Computational Statistics and Data Analysis* 71 (2014), pp. 643–651
52. White, G., Mazerolle, L., Porter, M., and Chalk, P. "Modelling the effectiveness of counter-terrorism interventions". In: *Trends and Issues in Crime and Criminal Justice* 457 (2014), pp. 1–8
53. White, G., Porter, M., and Mazerolle, L. "Terrorism Risk, Resilience, and Volatility: A Comparison of Terrorism in Three Southeast Asian Countries". In: *Journal of Quantitative Criminology* 29.2 (2013), pp. 295–320

54. Reich, B. J. and Porter, M. D. "Discussion of Estimating the historical and future probabilities of large terrorist events". In: *The Annals of Applied Statistics* 7.4 (2013), pp. 1871–1875
55. Porter, M. and White, G. "Self-exciting hurdle models for terrorist activity". In: *The Annals of Applied Statistics* 6.1 (2012), pp. 106–124
56. Porter, M. D. and Reich, B. J. "Evaluating temporally weighted kernel density methods for predicting the next event location in a series". In: *Annals of GIS* 18.3 (2012), pp. 225–240
57. Porter, M. D., White, G., and Mazerolle, L. "Evidence-Based Counterterrorism Policy". In: *Evidence-Based Counterterrorism Policy*. Ed. by Lum, C. and Kennedy, L. W. Springer New York, 2012. Chap. Innovative Methods for Terrorism and Counterterrorism Data, pp. 91–112
58. Porter, M. and Smith, R. "Network Neighborhood Analysis". In: *IEEE Int. Conf. on Intelligence and Security Informatics (ISI)*. Vancouver, B.C., 2010, pp. 31–36
59. Neimi, J., Porter, M., and Reich, B. "Mixture Likelihood Ratio Scan Statistic for Disease Surveillance". In: *Advances in Disease Surveillance* 5 (2008), p. 1
60. Porter, M. D. and Brown, D. E. "Detecting local regions of change in high-dimensional criminal or terrorist point processes". In: *Computational Statistics & Data Analysis* 51.5 (2007), pp. 2753–2768

In Progress

61. White, G. and Porter, M. "Contagion and Diffusion Models for the Dynamics of Terrorist Activity". Under Contract with CRC Press
62. Li, H. and Porter, M. D. "A Fast Two Stage Anomaly Detection Method for Large Dynamic Networks"
63. Liao, Y. and Porter, M. D. "The Predictability of Highway Crash Hotspots"
64. Posey, C., Porter, M., Lowry, P., and Moody, G. "Predictive Based Model Selection for Detecting Insider Cyber Security Threats"

Other

- 2020 Porter, M.D. "Don't Come Around Here No More or Party Like it's 1999?", Blog: Data Science, Statistics, and COVID-19. mdporter.github.io/blog/posts/2020-12-31-covid-event-risk/
- Porter, M.D. "To SIR, with Love", Blog: Data Science, Statistics, and COVID-19. mdporter.github.io/blog/posts/2020-05-12-to-sir-with-love/
- 2019 Porter, M.D. and Wang, K. (2019) "Bayesian Clustering using Non-negative Matrix Factorization", Proceedings of the 12th Scientific Meeting of the Classification and Data Analysis Group (CLADAG).
- 2017 Porter, M.D. (2017) "Multivariate Hawkes Point Process Models for Social Systems", Proceedings of the 62nd World Statistics Congress of the International Statistical Institute.
- 2016 Porter, M.D., Das T., and Zhang, R. (2016) Predictive Crash Analytics, Technical Report, CAPS (Center for Advanced Public Safety), University of Virginia.
- 2015 Porter, M.D. and Reich, B.J. (2015) crimelinkage: Statistical Methods for Crime Series Linkage, CRAN. (Open source software and user manual).
- 2014 Porter, M.D. and Reich, B.J. (2014) "Statistical Methods for Crime Series Linkage". The National Institute of Justice Technical Report.
- 2011 White, G., Porter, M.D. and Mazerolle, L. (2011). The Volatility and Risk of Terrorist Incidents: A Comparative Study Using Self-Exciting Models. Chapter 5 In *Modeling the Effectiveness of Counter-Terrorism Strategies in Indonesia, the Philippines, and Thailand*. A START/CEPS research report.

White, G., Mazerolle, L., and Porter, M.D. (2011). Modeling Counter-Terrorism Interventions and their Relative Effectiveness. Chapter 7 In *Modeling the Effectiveness of Counter-Terrorism Strategies in Indonesia, the Philippines, and Thailand*. A START/CEPS research report.

Porter, M.D. and Holland, B. (2011). Discovering Temporal Association between Two Types of Events. GeoEye Analytics Technical White Paper.

2010 Porter, M.D. (2010). Multivariate Change Detection. GeoEye Analytics Technical White Paper.

2009 Ghosh, S.K. and Porter, M.D. (2009). Spatio-Temporal Modeling of Health Related Incidents. Report to Centers for Disease Control and Prevention (CDC).

Dalton, J.R. and Porter, M.D. (2009). Geospatial Preference Models in Signature Analyst. SPADAC Technical White Paper.

Porter, M.D. (2009). Modeling Site Selection Behavior: Applications to Crime and Terrorism. In JSM Proceedings. Alexandria, VA: American Statistical Association, 5284-5293.

2007 Porter, M.D. (2007). Probability Estimates for Footwear Evidence.

2006 Porter, M.D. (2006). Detecting Space-Time Anomalies in Point Process Models of Intelligent Site Selection. Doctoral Dissertation, University of Virginia. Dissertations & Theses University of Virginia database. (Publication No. AAT 3225959).

FUNDING

2024–2029	\$1.1M	Agency for Healthcare Research and Quality (AHRQ) (co-PI) <i>Improving Pediatric Donor Heart Utilization with Predictive Analytics</i>
2024–2027	\$1.2M	National Institutes of Health (NIH) (co-I) <i>Research Experiences in Genomics Applications of Responsible Data Science</i>
2022–2024	\$133,078	Jefferson Trust (co-PI) <i>Optimizing Pediatric Donor Heart Utilization using Big Data Analytics</i>
2021–2023	\$350,000	National Cooperative Highway Research Program (NCHRP) (co-PI) <i>Implementing and Leveraging Machine Learning at State Departments of Transportation</i> *contracting issues prevented an award being issued
2020–2021	\$60,583	Engineering-in-Medicine seed grant (co-PI) <i>Impact Quantification of Donor Echocardiographic Data on Pediatric Heart Transplant Recipient Outcomes</i>
2021–2022	\$22,133	iThrive Pilot Translation and Clinical Studies (co-PI) <i>Impact Quantification of Donor Echocardiographic Data on Pediatric Heart Transplant Recipient Outcomes</i>
2020–2021	\$100,000	Ivy Foundation COVID-19 Translational Research Fund (I) <i>Epidemiologic Modeling, Public Health Surveillance and Sewershed Monitoring to Predict Surges in the COVID-19 Pandemic</i>
2020–2021	\$78,555	Vibrent Health (co-I) <i>Risk Prediction for COVID-19</i>
2020–2021	\$122,140	Engineering in Medicine Seed Grant (co-PI) <i>Deep Learning for Automated Identification and Classification of Gastrointestinal Landmarks in Video Capsule Endoscopies</i>
2016–2018	\$119,996	Center for Advanced Public Safety , University of Alabama (PI) <i>Predictive Crash Analytics</i>
2016–2017	\$20,000	Alabama Medicaid

		(Advisor) <i>Establishing a baseline distribution of risk scores for provider up-coding and fraud detection</i>
2015–2016	\$15,000	Culverhouse College of Business , University of Alabama <i>Grant Incubator Program</i>
2011–2013	\$551,656	National Institute of Justice (PI) <i>Statistical Methods for Spatio-Temporal Crime Series Linkage</i>
2007–2008	\$50,000	Centers for Disease Control and Prevention (co-PI) <i>Anomaly Detection in Space and Time</i>

SOFTWARE

R	crimelinkage : a set of tools to help crime analysts and researchers with tasks related to crime linkage. This package includes methods for criminal case linkage, crime series identification and clustering, and suspect identification.
RShiny	COVID-19 Event Risk Assessment Dashboard : an interactive tool that provides the estimated probability that someone at an event will unknowingly be contagious with COVID-19 based on the number of guests and their locations.

CONFERENCE PRESENTATIONS

2024	“Modeling contagion, excitation, and social influence with Hawkes point processes”. Advanced Studies Institute in Mathematics of Data Science & Machine Learning, Jan 8, Ugench State University.
2023	“Bridging the Gap in Pediatric Heart Transplants with Data Science”. Datapalooza, Nov 10, University of Virginia. “Forecasting Pediatric Heart Donors”. International Symposium of Forecasting, June 25-28, Charlottesville, VA. “Data Science Concepts in Risk Modeling”. National Institute of Justice (NIJ) National Research Conference, May 23-25, Arlington, VA.
2020	“A Tale of Two Metrics: Polling and Financial Contributions as a Measure of Campaign Performance”. 2020 Conference on Computational Sociology, Aug 7, Stanford University.
2019	“Detecting, identifying, and localizing radiological material in urban environments using scan statistics”. 2019 IEEE International Symposium on Technologies for Homeland Security, Nov 5 Nov 6, Woburn, MA. “Bayesian Clustering using Non-negative Matrix Factorization”. 12th Scientific Meeting of the Classification and Data Analysis Group (CLADAG), Sep 11 Sep 13, Cassino, Italy. “Spatial event hotspot prediction using multivariate Hawkes features”. Spatial Statistics 2019, Jul 10 Jul 13, Sitges, Spain. “Exciting Adventures in Crime Linkage”. Mathematical Criminology and Security Workshop, Mar 17 Mar 22, BIRS Banff, AB.
2018	“Predicting crime hotspots using rotational grids and sparse logistic regression models”. Joint Statistical Meetings, Jul 28 Aug 2, Vancouver, BC.
2017	“Understanding and Evaluating Predictive Crash Models”. SAMSI Summer Program on Transportation Statistics, Aug 14-18, Duke University. “Point Process Models for Social Systems”. ISI World Statistics Congresses, Jul 16-21, Marrakesh, Morocco.
2016	“Crime Linkage with Self-Exciting Point Process Models”. Joint Statistical Meetings, Jul 30 Aug 4, Chicago, IL.
2015	“Statistical Methods for Crime Series Linkage”. International Association of Crime Analysts Training Conference (IACA), Sept 20-25, Denver, CO.

- "A Statistical Approach to Crime Linkage". Joint Statistical Meetings, Aug 8 - 13, Seattle, WA.
Media Coverage: (i) <http://www.forensimag.com/news/2015/08/not-just-hunch-new-software-uses-stats-solve-predict-burglary-experts-say> (ii) Software identifies missing link, Police Professional, (20 Jan 2016)
- 2014 "Interval-Censoring Methods for Aoristic Crime Analysis". Joint Statistical Meetings, Aug 2- 7, Boston, MA.
- 2013 "An Interval Censoring Approach for Aoristic Analysis". American Society of Criminology Annual Meeting (ASC) Nov 19-23, Atlanta, GA.
"Statistical Crime Linkage: Case Linkage, Crime Series Identification, and Clustering". International Association of Crime Analysts Training Conference (IACA), Sept 9-13, Fort Lauderdale, FL.
"Discussion of Clauset and Woodard: Estimating the Historical and Future Probabilities of Large Terrorist Events". Joint Statistical Meetings, Aug 3- 8, Montreal, Canada. (Invited Discussant for "Annals of Applied Statistics" session.)
- 2012 "Identifying suspects, discovering new offenders and linking crimes: Crime series identification using model-based clustering". American Society of Criminology Annual Meeting (ASC) Nov 14-17, Chicago, IL.
"Self-Exciting Hurdle Models for Terrorist Activity". Joint Statistical Meetings, Jul 28-Aug 2, San Diego, CA. (Invited "Best of the Annals of Applied Statistics" session.)
"Weighted Kernel Density for Predicting the Location of the Next Event in a Series". Quantitative Methods in Defense and National Security (QMDNS), May 1, George Mason University
- 2011 "Bayesian Crime Series Linkage Analysis". American Society of Criminology Annual Meeting (ASC) Nov 16-19, Washington, D.C.
"Bayesian Crime Series Linkage Analysis". NIJ Crime Mapping Conference (MAPS), Oct 19-21, Miami, FL.
"StoNA: Structure to Network Activity". HSCB Focus 2011, Feb 8-10, Chantilly, VA. (Dalton, J., Porter, M.D., Abrams, M., and Valore, J.)
- 2010 "Self-exciting Hurdle Models for Terrorism". AMS Fall Western Section Meeting, Oct 9-10, Los Angeles, CA.
"Is Terrorism Contagious? Modeling Indonesian Terrorism with Self-Exciting Hurdle Models". Joint Statistical Meetings, Aug. 1-5, Vancouver, BC.
"Discovering Terrorist Subgroups with Network Neighborhood Analysis". Quantitative Methods in Defense and National Security, May 25-26, George Mason University.
"Network Neighborhood Analysis". IEEE Intelligence and Security Informatics, May 23-26, Vancouver, BC. (Smith, R. and Porter, M.D.)
- 2009 "Modeling Site Selection Behavior: Applications to Crime and Terrorism". Joint Statistical Meetings, Aug. 1-6, Washington, D.C.
- 2008 "Mixture Likelihood Ratio Scan Statistic for Disease Outbreak Detection". International Society for Disease Surveillance (ISDS) annual conference, Dec 3-5, Raleigh, NC. (Neimi, J.B., Porter, M.D., and Reich, B.J.)
"Probability Estimates for Footwear Evidence". NC IAI Fall Conference, October 2-3, Atlantic Beach, NC.
"Probability Estimates for Footwear Evidence". International Association for Identification 93rd International Educational Conference, August 17-23, Louisville, KY.
"A Martingale Methodology for the Quick Identification of Point Process Anomalies". INTERFACE 2008, May 21-24, Durham, NC.
- 2007 "An Adaptive Methodology for the Quick Identification of Space-Time Anomalies". INFORMS annual meeting, Nov 4-7, Seattle, WA.
"The Search for Spatial and Spatio-Temporal Anomalies". Army Conference on Applied Statistics, Oct 17-19, Rice University.

- “Some adaptive approaches for space-time anomaly detection”. First International Workshop in Sequential Methodologies, July 22-25, Auburn University.
- “Change detection for space-time Poisson point processes with applications to defense and homeland security”. Spring Research Conference on Statistics in Industry and Technology, May 21-23, Iowa State University.
- “Anomaly Detection in Space-Time Point Processes”. 32nd Spring Lecture Series, Spatial and Spatio-Temporal Statistics, Apr. 12-14, University of Arkansas.
- “Anomaly Detection in Space-Time (and higher dimensional) Point Processes”. Quantitative Methods in Defense and National Security, Feb. 7-8, George Mason University.
- 2006 “Modeling the Interaction between Intelligent Site Selection and Other Stochastic Processes with Applications to Terrorism”. Joint Statistical Meetings, Aug. 6-10, Seattle, WA. (Robinson, C.D., Porter, M.D., and Brown, D.E.)
- 2005 “Feature Selection, Prediction, and Change Detection in Terrorist/Insurgency Processes Using a Spatial Point Process Approach”. Knowledge Fusion Research Workshop, Nov. 29- Dec. 1, Annapolis, MD.
- “Detecting Changes in Criminal or Terrorist Processes”. SAMSI National Defense and Homeland Security Kickoff Workshop, Sept. 11-14, RTP, NC.
- “Finding Changing Crime Regions: Use of High Dimensional Geographic Feature Space and Classification Trees”. Proceedings of the Eighth Crime Mapping Research Conference, Sept. 7-10, Savannah, GA.
- 2003 “A Modified Response Surface Methodology for Knowledge Discovery: Optimization via Functional Approximation”. Poster Presentation. Winter Simulation Conference, Dec. 7-10, New Orleans, LA.
- “Obsolescence Management Decision Making in a Life-Cycle Extension”. Proceedings of Third COG Conference. May 19-22, 2003, Glasgow, Scotland.
- 2002 “Functional Discovery to Enable Confident Change”. Proceedings of Fifteenth International Conference on Systems Engineering. August 6-8, 2002, Las Vegas, NV. pp. 273-277. (Tomlinson, J., Porter, M.D., and Mahaffey, W.R.)

OTHER TALKS

- 2021 Panelist at NIJ Recidivism Forecasting Challenge Symposium. Dec 1-2.
Modeling contagion, excitation, and social influence with Hawkes point processes. Amazon Tech Talk, Dec 2021.
- 2020 *To SIR, with love*. Elder Research, Tech Talk, May 29, 2020.
- 2018 *Modeling contagion, excitation, and social influence with Hawkes point processes*. University of Virginia, Department of Statistics, Nov 30, 2018.
Modeling contagion, excitation, and social influence with Hawkes point processes. University of Virginia, Data Science Institute, April 17, 2018.
- 2015 *Self-exciting Hurdle Models for Terrorist Activity*. Mississippi State University, Statistics Seminar, April 7, 2015.
- 2014 *Predictive Crime Analytics*. Cambridge, MA, Aug 11-12 2014.
- 2013 *Statistical Crime Series Linkage*. The University of Alabama, Applied Statistics Seminar, Sept 30, 2013.
Self-exciting Hurdle Models for Terrorist Activity. The University of Alabama, Applied Statistics Seminar, Feb 20, 2013.
Analyzing Terrorist Attack Patterns with Self-exciting Models. Daniel Rose Yale-Technion Counterterror Research Initiative Seminar, Feb 26, 2013 Yale University.
- 2011 *Self-exciting Hurdle Models for Terrorism: Understanding the Risk of Indonesian Terrorist Attacks*. University of Louisville, Mathematics Colloquium, Mar 4, 2011.
- 2008 *Regression and Least Squares: A Matlab Tutorial*. SAMSI/CRSC Undergraduate Workshop, N.C. State University, Raleigh, NC., May 19-23, 2008.

	<i>Cluster Detection in Geographic Sensor Networks</i> . SAMSI Undergraduate Workshop, RTP, NC. February 29-March 1, 2008.
2007	<i>Intelligent Site Selection Models for Asymmetric Threat Prediction and Decision Making</i> . SAMSI Undergraduate Workshop, Durham, NC, November 9-10, 2007.
	<i>The Search for Spatial and Spatio-Temporal Anomalies</i> . Virginia Tech, Department of Statistics, August 30, 2007.
	<i>Which targets are next? Modeling Intelligent Site Selection with Applications to Terrorism</i> . Applications of Statistical Modeling and Simulation, University of Delaware, April 27, 2007.
	<i>Anomaly Detection in Space-Time (and higher dimensional) Point Processes</i> . North Carolina State University, Department of Statistics, February 22, 2007.
	<i>Intelligent Site Selection and Anomaly Detection</i> . ADAC Lab, North Carolina State University, Department of Electrical and Computer Engineering, February 15, 2007.
2006	<i>Anomaly Detection in Space-Time (and higher dimensional) Point Processes</i> . University of South Carolina, Department of Statistics, October 26, 2006.
	<i>Anomaly Detection in Space-Time Point Processes</i> . Ninth Meeting of New Researchers in Statistics and Probability, August 1-5, 2006, Seattle, WA.
	<i>Using Marked Spatial Point Process Models to Detect Change in Intelligent Site Selection Processes</i> . University at Buffalo, Department of Systems and Industrial Engineering, April 3, 2006.
	<i>Using Point Process Residuals in Space-Time Anomaly Detection</i> . SAMSI Anomaly Detection Working Group Meeting, March 23, 2006.

TEACHING

University of Virginia

DS 6410	<i>Machine Learning II: Methods & Application</i> : Sp 23, Sp 24, Sp 25
DS 6030	<i>Statistical Learning</i> : Fa 21, Fa 22, Fa 23, Fa 24
SYS 6018	<i>Data Mining</i> : Sp 19, Fa 19, Su 20, Fa 20, Sp 21, Sp 22, Sp 23, Sp 24, Sp 25
SYS 4055	<i>Systems Engineering Design Colloquium II</i> : Fa 18
SYS 4053	<i>Systems Design (Capstone)</i> : 18-19, 19-20, 20-21, 23-24
SYS 3055	<i>Systems Engineering Design Colloquium I</i> : Fa 18
SYS 2202	<i>Data and Information Engineering</i> : Fa 19
GBUS 7600	<i>Data Analysis and Optimization</i> : Sp 19

University of Alabama

ST 697	<i>Advanced Statistical Learning</i> : Sp 14, Fa 15, Fa 17
ST 597	<i>Introduction to Data Analytics</i> : Sp 15, Sp 16, Sp 17
ST 591	<i>Introduction to Statistical Learning</i> : Fa 16, Su 17
ST 560	<i>Statistical Methods</i> : Fa 15, Fa 16
ST 554	<i>Mathematical Statistics I</i> : Fa 16, Fa 17
ST 260	<i>Statistical Data Analysis</i> : Fa 13, Sp 14, Fa 14, Sp 15, Sp 16, Sp 18

North Carolina State University

ST 516	<i>Experimental Statistics for Engineers II</i> : Sp 08
ST 515	<i>Experimental Statistics for Engineers I</i> : Fa 06, 07
ST 371	<i>Probability and Distribution Theory</i> : Sp 07

Student Supervision

2024	Jiahao Tian (PhD Data Science, University of Virginia)
2024	Julia Sharff (MS Systems & Information Engr., University of Virginia)
2023	Moeen Mostafavi (PhD Systems & Information Engr., University of Virginia)
2023	Yichen Jian (PhD Systems & Information Engr., University of Virginia)
2017	Ketong Wang (PhD Statistics, University of Alabama)

SERVICE ACTIVITIES

2024	Chair <i>Forecasting for Social Good</i> section of International Institute of Forecasting (IFF)
2024	Chair AGF Search Committee, School of Data Science, University of Virginia
2023	Program and Organizing committee <i>International Symposium of Forecasting</i> , June 2023, Charlottesville, VA.
2022-2024	Co-chair <i>Forecasting for Social Good</i> section of International Institute of Forecasting (IFF)
2022+	Steering Committee UVA Collaboratory for Applied Data Science in Business
2022, 2024	Guest editor International Journal of Forecasting special edition: <i>Forecasting for Social Good</i>
2021	Technical Program Committee, IEEE Big Data 2021 Workshop on Big Data Analytics for COVID-19 (BDA COVID-2021)
2021	Technical Program Committee, IEEE Big Data 2021 Workshop on Multimodal Big Data (MMBD 2021)
2020	Technical Program Committee, IEEE Big Data Workshop on Data Science for Smart and Connected Communities
2019-2021	Systems Engineering Undergrad Committee, Engineering Systems & Environment Department
2019-2022	Partnership Chair, SIEDS Steering Committee
2020-2021	SDS Admissions Committee
2019-2021	SDS Academic Affairs Committee
2019-2021	SDS Graduate Education Committee
2019-2021	General Chair, Forecasting for Social Good Workshop
2020	Distinguished Fellow Review Committee, School of Engineering and Applied Science
2019	Task force on Colloquium, Engineering Systems & Environment Department
2017-2018	Guest editor International Journal of Forecasting special edition: <i>Forecasting for Social Good</i>
2016-2018	Technology Research Advisory Committee, University of Alabama
2015-2018	Medicaid Research Committee, Culverhouse College of Commerce
2016-2018	Business Analytics Steering Committee, Culverhouse College of Commerce
2014-2016	MS-Business Analytics Committee, ISM Dept, University of Alabama
2013-2016	Research Advisory Committee, University of Alabama
2013-2014	Secretary/Treasurer of the American Statistical Association Section on Defense and National Security (ASA-SDNS)
2013-2019	Grant reviewer for National Institute of Justice (NIJ) Graduate Research Fellowship (2013, 2016, 2019)
2009-2010	Publications Officer of the American Statistical Association Section on Defense and National Security (ASA-SDNS)
2007-2012	Program committee for Quantitative Methods in Defense and National Security (QMDNS) conference (2007, 2010, 2012)

2005+	Article reviews for JASA, CSDA, JABES, AOAS, International Journal of Forecasting, Biostatistics, Biometrics, Journal of Applied Statistics, Management Science, Criminology, Legal and Criminological Psychology, Crime Mapping, Journal of Experimental Criminology, Journal of Quantitative Criminology, Journal of Offender Therapy and Comparative Criminology, NetCrime, Journal of Forensic Science, Journal of Criminal Justice, Crime Science, IEEE Transactions on SMC, Mathematical and Computer Modeling, Forest Science, Annals of GIS, Applied Geography, Int. J. of Information Technology and Decision Making, Decision Sciences, Homeland Security Technologies.
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OTHER ACTIVITIES

2014	Statistical Training: Predictive Crime Analytics Workshop, Aug 11-12, Cambridge, MA. Led two-day workshop on geospatial predictive methods for crime analysts.
2007-2008	Director of VIGRE Undergraduate Research Project: Examining Footwear Evidence for Identification
2008	Session organizer "Spatial Risk Mapping: Prediction and Change Detection", Interface 2008, Durham, NC, May 2008.
2007	Attended IPAM Crime Hot Spots: Behavioral, Computational and Mathematical Models
2006	Session organizer "Prediction and Detection in Defense and Homeland Security Applications", Joint Statistical Meetings Seattle, WA, August 2006

AWARDS

2021	Outstanding Graduate Teaching in Systems Engineering, University of Virginia
2007	Louis T. Rader Outstanding Ph.D. Student Award, awarded by the Department of Systems and Information Engineering, University of Virginia
2006	Winner of the Student Paper Competition for ASA Section on Statistics in Defense and National Security (ASA-SDNS)
2005	NSF travel award for NATO ASI (Advanced Study Institute): Multisensor Data and Information Processing for Rapid and Robust Situation and Threat Assessment
2005	Travel award for the Eighth Crime Mapping Research Conference, Savannah, GA.
2005	Travel award for the SAMSI National Defense and Homeland Security Kickoff Workshop, RTP, NC.